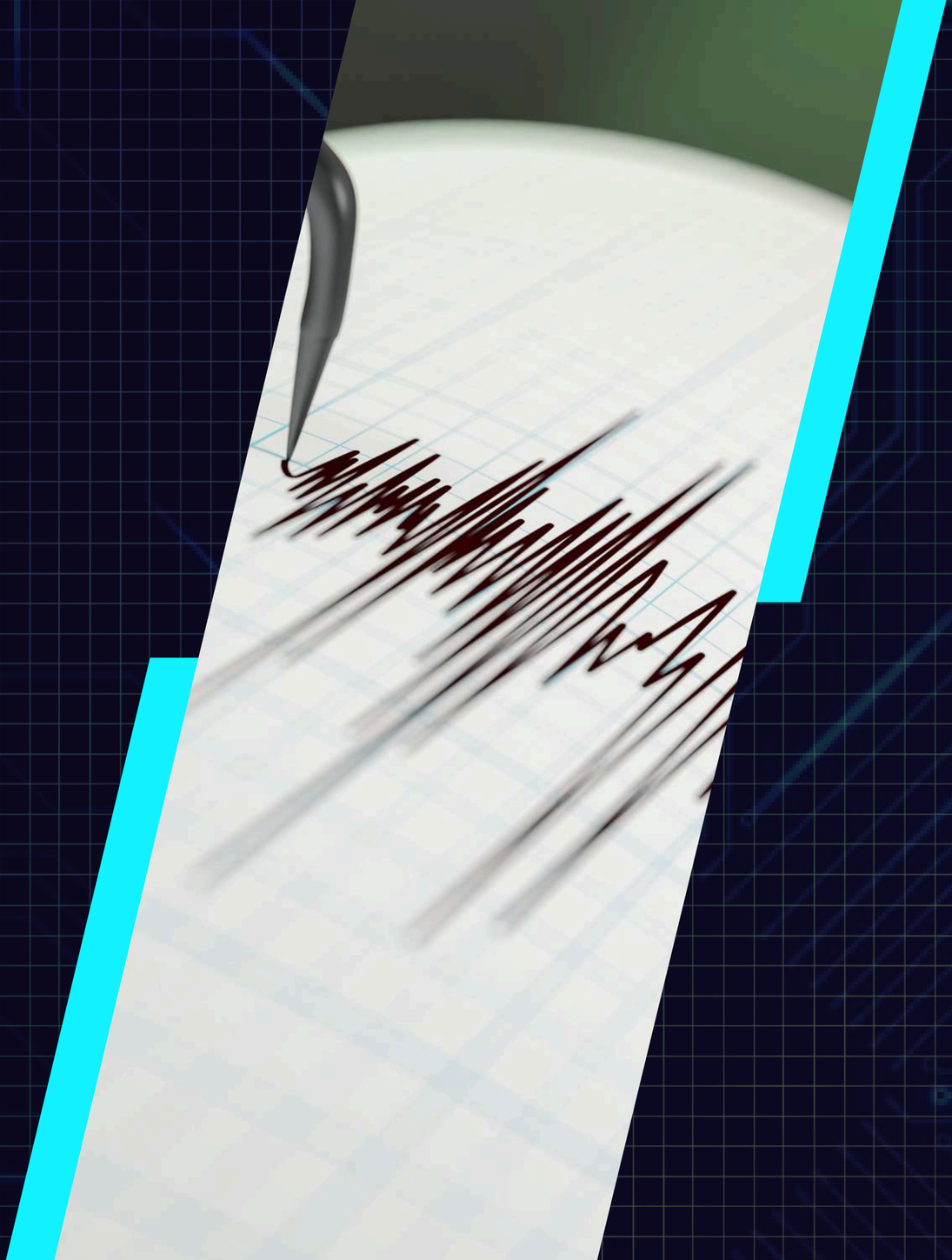
EARTHQUAKE ALERT SYSTEM

By

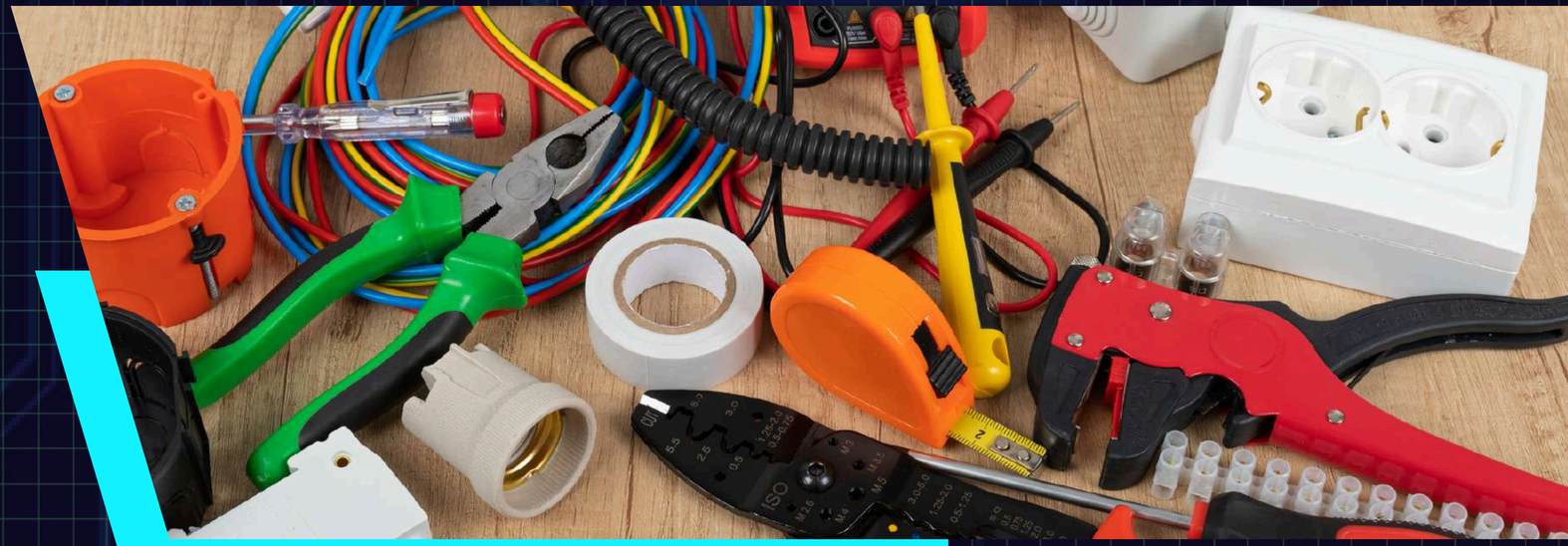
SHREE DHARSHAN S
SUGI SIVAM S



INTRODUCTION

- Earthquakes are sudden and unpredictable natural disasters that cause major damage to life and property.
- In regions where earthquakes are common, early detection systems can significantly reduce the impact by issuing timely alerts.
- This project proposes an Arduino-based earthquake alert system using an MPU6050 accelerometer and gyroscope sensor.
- The system detects vibrations and triggers an alarm (buzzer) and LED alert when the vibration exceeds a certain threshold.

COMPONENTS REQUIRED



01 MPU6050 Sensor

02 Arduino UNO

03 Buzzer

04 Bread Board

05 LED

06 Jumper Wires

07 1K Resistor

CODE

```ino
#include <Wire.h>
#include <MPU6050_light.h>

MPU6050 mpu(Wire);

unsigned long timer = 0;
int threshold = 2;  // Sensitivity for detecting movement
int ledPin = 8;     // LED Pin
int buzzerPin = 9;  // Buzzer Pin

void setup() {
  pinMode(ledPin, OUTPUT);
  pinMode(buzzerPin, OUTPUT);

  Serial.begin(9600);
  Wire.begin();

  byte status = mpu.begin();
  Serial.print("MPU6050 status: ");
  Serial.println(status);
  while (status != 0) {
    Serial.println("MPU6050 connection failed. Check wiring.");
    delay(1000);
  }

  Serial.println("Calibrating...");
  mpu.calcOffsets();  // Takes ~5 seconds
  Serial.println("Done calibrating ✅");
}

void loop() {
  mpu.update();
```

```ino
int accX = mpu.getAccX();
int accY = mpu.getAccY();
int accZ = mpu.getAccZ();

float magnitude = sqrt(accX * accX + accY * accY + accZ * accZ);  // Combined magnitude of acceleration

// Debugging: Print magnitude to check the sensor output
Serial.print("Magnitude: ");
Serial.println(magnitude);

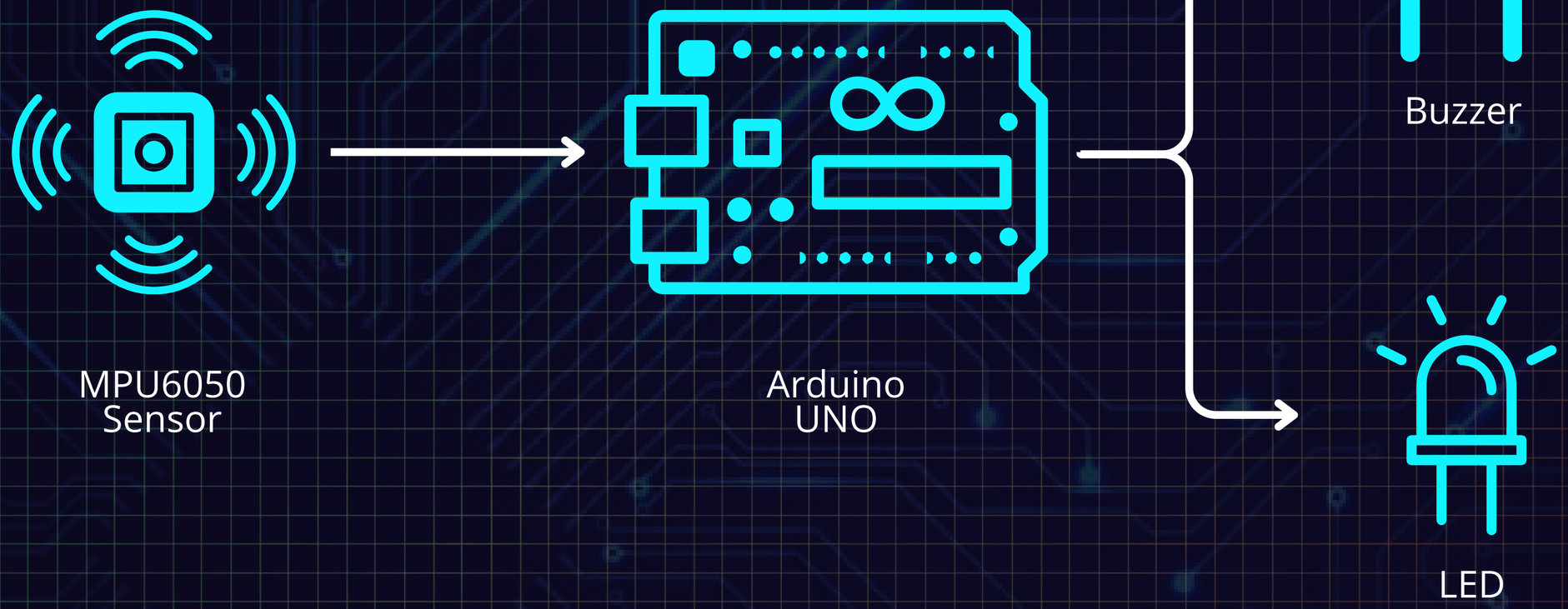
if (magnitude > threshold) {
  unsigned long currentMillis = millis();
  if (currentMillis - timer > 500) {  // Ensure motion lasts for at least 500ms
    Serial.println("Vibration detected! Triggering alert...");

    digitalWrite(ledPin, HIGH);    // Turn on LED
    digitalWrite(buzzerPin, HIGH); // Turn on Buzzer
    delay(1000);                   // Keep alert for 1 second
    digitalWrite(ledPin, LOW);     // Turn off LED
    digitalWrite(buzzerPin, LOW);  // Turn off Buzzer
    timer = currentMillis;         // Reset timer
  }
}

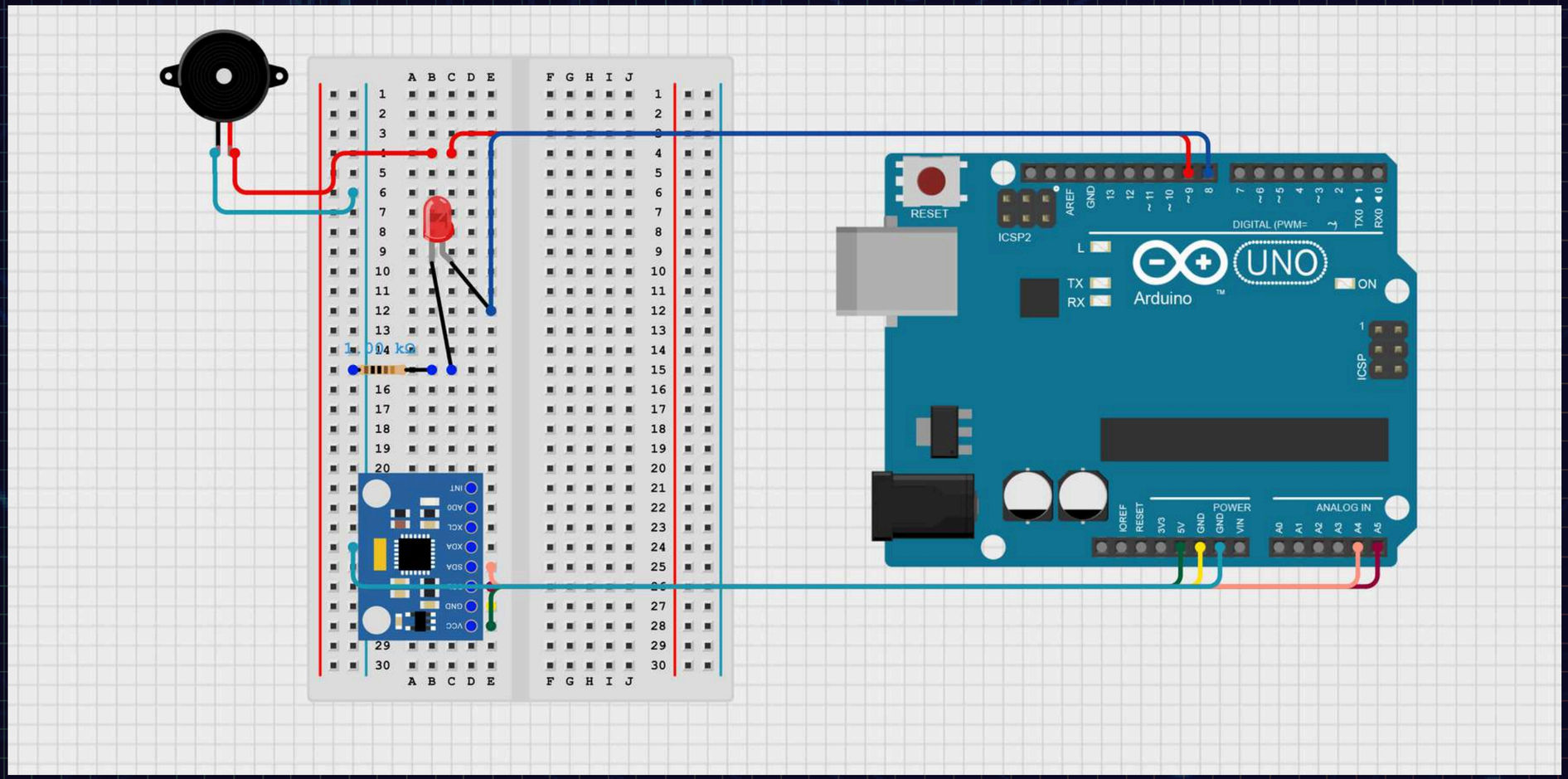
// Print the accelerometer data to serial monitor
Serial.print("Acc X: "); Serial.print(accX);
Serial.print(" | Acc Y: "); Serial.print(accY);
Serial.print(" | Acc Z: "); Serial.println(accZ);

delay(700);  // Delay to slow down the data output in the Serial Monitor (200 ms)
```

BLOCK DIAGRAM



CIRCUIT DIAGRAM



FUTURE SCOPE

INTEGRATION WITH IOT

Send real-time earthquake alerts via SMS or email using Wi-Fi/Bluetooth modules.

DATA LOGGING

Store vibration data for further analysis and pattern detection.

MOBILE APP ALERT

Notify users through mobile app integration for quick alerts.

USE OF GPS MODULE

Track and log the location of the earthquake detection.



THANK YOU

